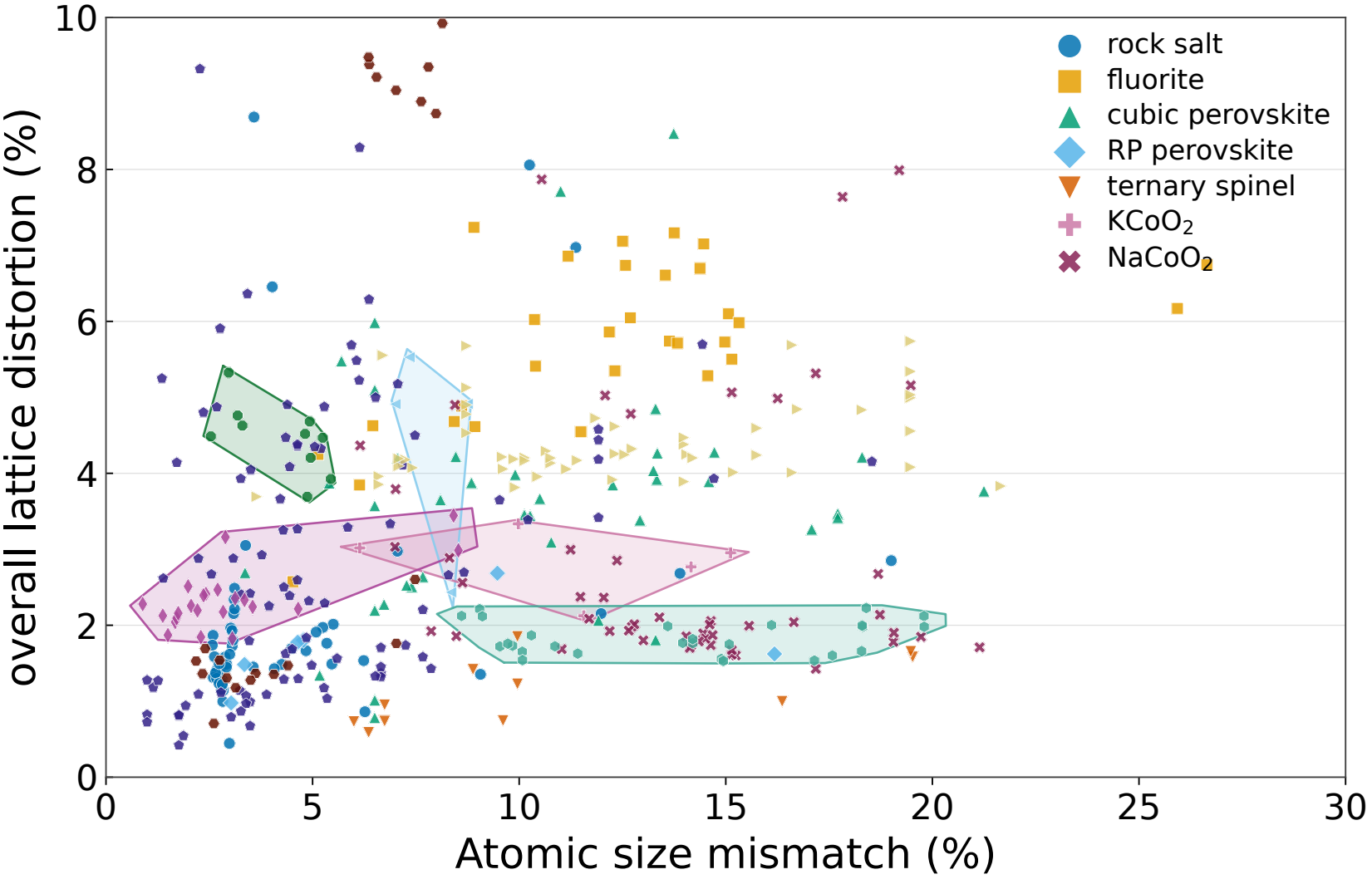
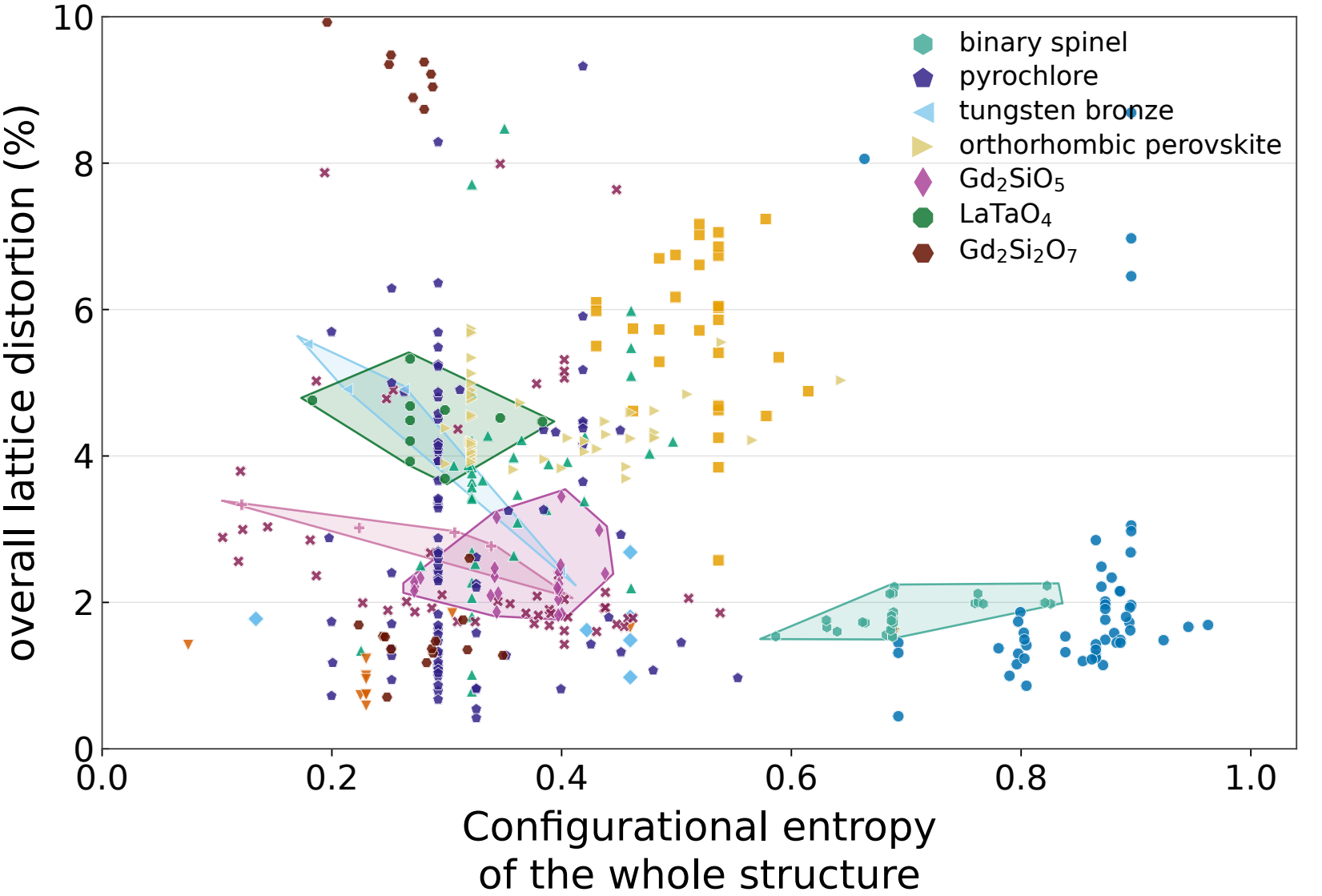
(a)


overall lattice distortion (%)
Atomic size mismatch (%)
0
2
4
6
8
10
0
5
10
15
20
25
30
rock salt
fluorite
cubic perovskite
RP perovskite
ternary spinel
$KCoO_2$
$NaCoO_2$

(b)


overall lattice distortion (%)
Configurational entropy
of the whole structure
0
2
4
6
8
10
0.0
0.2
0.4
0.6
0.8
1.0
binary spinel
pyrochlore
tungsten bronze
orthorhombic perovskite
Gd_2SiO_5
$LaTaO_4$
$Gd_2Si_2O_7$